



HostelHub

Complete Application Architecture & Feature Master Audit

DOCUMENT SCOPE

Full Panel-Wise Audit, Data Storage & Security Breakdown

PRIMARY TECH STACK

Node.js, Express, SQLite, Sequelize, Angular, Socket.io

SYSTEM VERSION

HostelHub v2.4 Enterprise Edition

LANGUAGE & FORMAT

Hinglish • Executive Master PDF Report

1. 📱 HostelHub Ecosystem Overview (Full App Summary)

HostelHub ek multi-role, end-to-end Smart Hostel Management Platform hai jo student residence lifecycle ko completely digitize karta hai. System 4 dedicated frontends (Portals) aur 1 centralized backend service par chalta hai:

🎓 Student Portal (frontend-user)

Complaint lodging, live roll call attendance, mess feedback, real-time chat, profile management, and push alerts.

👮 Warden Portal (frontend-warden)

Block-wise roll call marking, complaint resolution & staff task assignment, mess menu management, announcements broadcast.

🔧 Staff / Technician Portal (frontend-staff)

Task execution dashboard for Electricians, Plumbers & IT support, task status updates, completion proof upload.

🔑 Super Admin Portal (frontend-admin)

User provisioning, student account approvals, hostel block & room capacity management, global app settings.

2. Panel-Wise Complete Features Audit

PANEL 1



Student Portal (User Dashboard)

- **Authentication & Onboarding:** Email/Password login, Google OAuth 2.0 Single Sign-On, Student Registration form, Password Reset via Email OTP.
- **Profile Management:** View & edit personal details, Profile Picture upload (Multer integration), Room Number, Hostel Block, Roll Number, Bio, Gender, and Batch info.
- **Complaint Lodging Engine:** Raise complaints under categories (Plumbing, Electrical, Wi-Fi, Food Quality, Misconduct). Upload photo evidence directly from phone camera/gallery. Real-time status tracking (Pending → In Progress → Resolved). Chat/comments on specific complaints.
- **Attendance History:** View monthly attendance calendar, attendance status badges (Present, Absent, Outstation Leave).
- **Mess Feedback & Daily Menu:** View today's and weekly Mess Menu (Breakfast, Lunch, Snacks, Dinner). Submit 1-5 Star Ratings and comments for meal quality.
- **Real-Time Community & Chat:** Direct 1-on-1 messaging (Student-to-Warden, Student-to-Student), Hostel Block & Floor Group Chat rooms with real-time Socket.io instant messaging, unread badges, and attachment sharing.
- **Notifications & Circulars:** Live push alerts for complaint status changes, new announcements, and warden broadcasts.

PANEL 2

**Warden Portal (Operational Command)**

- **Operational Dashboard:** Block-wise student count, pending complaints widget, today's attendance percentage summary.
- **Daily Roll Call Attendance:** Mark daily student attendance (Present / Absent / Outstation), filter by floor or room, 1-click "Mark All Present" bulk feature.
- **Complaint Management & Staff Routing:** View complaints raised in their hostel block, inspect uploaded evidence photos, update status (In Progress, Resolved, Rejected), assign complaints to specific maintenance staff (e.g. Electrician Ramesh).
- **Mess Operations Oversight:** Edit & update weekly mess menu schedule, monitor daily student mess ratings and feedback comments.
- **Official Broadcast Circulars:** Publish official announcements/notices targeted to specific hostel blocks or entire college with push notifications.
- **Direct Student & Staff Chat:** Real-time messaging with students for issue resolution and staff for maintenance updates.

PANEL 3

**Staff / Maintenance Portal (Work Order Engine)**

- **Assigned Task Dashboard:** View maintenance work orders assigned by Wardens based on trade (Electrical, Plumbing, Wi-Fi/IT, Carpentry).
- **Task Progress Update:** Change task status from Pending → In Progress → Resolved. Add work completion notes.
- **Warden Coordination Chat:** Direct chat channel with the assigning Warden for room entry permissions or material required.

PANEL 4

**Super Admin Portal (System Management)**

- **User Access & Account Verification:** View master user directory across all roles ('student', 'warden', 'staff', 'admin'). Approve pending student registrations, block suspicious accounts, change user roles.
- **Hostel Infrastructure Management:** Create & edit Hostel Blocks (e.g., Boys Block A, Girls Block B), Floors, Rooms, Room sharing capacity (Single, Double, Triple), Room Amenities (AC / Non-AC). Assign rooms to students.
- **Complaint Categories & Settings:** Manage system-wide complaint categories, app parameters, and global system notices.
- **Centralized Master Reports:** Export student master database, attendance logs, and complaint resolution analytics.

3. Data Storage & Backend Architecture Engine

Backend Node.js + Express framework par structured Sequelize ORM ke sath chalta hai:

**Database Engine**

SQLite Database (database.sqlite) managed via Sequelize ORM with automatic table migrations & indexing.

**File Storage (Uploads)**

Local uploads/ directory managed by Multer middleware for profile pictures & complaint evidence images.

**Real-Time WebSockets**

Socket.io engine (socket.js) handling instant messaging, chat room joining, and real-time notification push.

**Database Migration Scripts**

Automated scripts (initDb.js, seed.js, migrate_add_google_id.js, migrate_add_chat_fields.js).

**Core Database Models & Schema Breakdown:**

- **User Model (User.js):** Stores user profiles, role enum (student, warden, staff, admin), status (active, pending_verification, blocked), Google ID, hashed password, room number, hostel block, roll number, phone, gender, and batch.
- **Attendance Model (Attendance.js):** Stores daily roll call entries linked to User ID, date, status (present, absent, outstation), and warden ID.
- **Complaint Model (Complaint.js):** Stores complaints with category, title, description, proof image URL, status (pending, in_progress, resolved, rejected), student ID, assigned staff ID, hostel block.

- **ChatMessage & GroupChat Models (ChatMessage.js, GroupChat.js):** Stores 1-on-1 and group messages, sender ID, receiver ID, group ID, message text, attachment URL, read receipts.
- **MessMenu, MessFeedback & MessSkip Models:** Stores day-wise menu schedules, 1-5 star meal feedback ratings, student comments, and skip requests.
- **Announcement & Notification Models:** Stores published circulars, targeted blocks, push notification logs, and read flags.
- **PasswordResetOTP Model:** Stores email-linked hashed OTP tokens and expiration timestamps.

4. Security, Privacy & Data Protection Architecture

SECURITY



Complete System Security Framework

- **JWT Token Authentication:** Stateless JSON Web Token authentication with HTTP Bearer Authorization header. Tokens carry user ID, email, and role claims with expiration enforcing re-authentication.
- **Role-Based Access Control (RBAC Middleware):** Strict `auth.js` middleware enforcing route protection (e.g. `authorizeRoles('admin', 'warden')`). Students cannot access Warden or Admin APIs.
- **Bcrypt Password Hashing:** All user passwords are salted & hashed using `bcryptjs` (10 salt rounds) before storing in DB. Plaintext passwords are never saved.
- **Google OAuth 2.0 Security:** Secure Google token validation for single sign-on without password exposure.
- **SQL Injection & XSS Protection:** Sequelize ORM provides fully parameterized SQL queries, eliminating SQL Injection vulnerabilities. Inputs are sanitized against XSS attacks.
- **Multer Storage Security:** File upload validation enforcing allowed MIME types (JPEG, PNG, WEBP), file size caps, and random filename generation to prevent arbitrary file execution scripts.
- **Environment Isolation:** Sensitive keys (JWT Secret, Database path, Server Port) isolated in server `.env` file.
- **Socket.io Room Authorization:** Real-time chat socket connections require handshake token authentication, isolating group chat channels to authorized members only.

5. Live Features & Native Mobile Apps (Things Often Missed)

- **Native Android APK Builds:** Web-to-Native WebView wrappers available in root directory:
 - `HostelHub-Latest.apk` (Student/Staff App)
 - `HostelHub-Admin-Latest.apk` (Warden/Admin Management App)
- **Adaptive Branding & Responsive UI:** Custom CSS glassmorphism theme with dark/light mode toggle, mobile navigation drawer, and desktop sidebar layout.
- **Real-time Unread Message Counters:** Dynamic badge notifications in chat tabs synced via Socket.io events.
- **Database Seeding & Recovery:** Automated database seed scripts (`seed.js`) for initial admin account and test data initialization.

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